

Insights Into Visible Light Positioning: Range Tracking and Bayesian Cramér-Rao Lower Bound Analysis

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Abstract—Visible Light Communication (VLC) is increasingly seen as a promising solution for indoor ranging systems. This study investigates user range estimation from a light source in a single-input single-output (SISO) system, considering input-signal-dependent shot noise (ISDSN). We apply extended Kalman filter (EKF), maximum likelihood (ML), and nonlinear least square (NLS) estimator techniques for range estimation and derive the Bayesian Cramér-Rao lower bound (BCRLB) to assess system performance. Monte Carlo simulations confirm that the proposed estimators align well with theoretical expectations. The results indicate that estimation error rises with higher ISDSN levels for all techniques and the BCRLB, with the EKF showing superior accuracy in range estimation compared to ML and NLS.

Index Terms—Visible light communications, range estimation, Bayesian Cramér-Rao lower bound, extended Kalman filter, input-signal-dependent shot noise.

I. INTRODUCTION

OVER time, visible light communication (VLC) has become a significant focus for indoor environments due to its high bandwidth capabilities, providing a secure and high-speed communication alternative [1]. Additionally, light-emitting diode (LED) based visible light positioning (VLP) holds the potential for precise indoor positioning by utilizing existing VLC infrastructure, which reduces costs [2].

In VLP systems, two main noise sources are thermal noise and signal-dependent (shot noise). Thermal noise results from electron excitation in the photodiode (PD), while shot noise depends on the input signal strength, and is termed input signal-dependent shot noise (ISDSN) [3], which significantly degrades VLP system performance [3].

The authors in [4] derived the Cramer-Rao lower bound (CRLB) for a 3D VLP system using received signal strength (RSS), time of arrival (TOA), and time difference of arrival (TDOA). Additionally, [5] combined RSS and angle of arrival (AOA) for a hybrid algorithm to enhance remote receiver positioning [4], [5]. The Kalman filter (KF) is widely used in VLP to enhance estimation accuracy [6]. However, many positioning systems are nonlinear, necessitating algorithms like the extended Kalman filter (EKF) to handle nonlinearity [8]. The Bayesian Cramer-Rao lower bound (BCRLB) is used

to evaluate estimation performance by incorporating prior stochastic information [7].

The main contributions of our work can be outlined as follows: 1) We address range estimation between a visible light transmitter and a PD receiver in a scenario affected by ISDSN using multiple techniques, including recursive and non-recursive methods. 2) For non-recursive techniques, we implement ML and nonlinear least squares (NLS) estimators and derive the closed-form expression for the NLS mean square error (MSE) through the parameter method transformation. 3) For the recursive technique, we utilize the EKF and compare its performance with that of the non-recursive techniques. Additionally, we investigate the impact of the receiver plane's tilting angle on the performance of EKF estimation. 4) We investigate the effectiveness of the EKF in mitigating the impact of ISDSN in VLP range estimation. 5) We derive and analyze the BCRLB, discussing the impact of ISDSN on the BCRLB.

II. SYSTEM MODEL

In our system setup, an LED is affixed to the ceiling at a predetermined height. We employ a single-input single-output (SISO) VLP system for ranging detection. The LED is responsible for covering a specific circular area with radius Y , as depicted in Fig.1 and the received signal at the PD can be written as [9] and [10]

$$r = hx + \sqrt{hxn_{ds}} + n, \quad (1)$$

where x , is the transmitted pilot (known) signal, n is the signal-independent thermal noise, $n \sim \mathcal{N}(0, \sigma_n^2)$, and $\sqrt{hxn_{ds}}$ denotes the ISDSN, where $n_{ds} \sim \mathcal{N}(0, \sigma_{ds}^2)$. We represent the ISDSN in terms of thermal noise as $n_{ds} \sim \mathcal{N}(0, \sigma_n^2 \zeta^2)$, where $\zeta^2 = \sigma_{ds}^2 / \sigma_n^2$, is the shot noise scaling factor that indicates the strength of the ISDSN compared to the thermal noise.

In the context of the term ζ^2 , it's important to highlight that this parameter can be set within a practical range, typically spanning from 1 to 10, by configuring the receiver's parameters [3], [9]. In general, shot noise typically follows a Poisson distribution. Nevertheless, as the number of received photons increases significantly, it converges towards a Gaussian distribution. This behavior is related to the high-intensity nature of light in VLC systems [3], [11]. The channel gain h between the LED and the user can be expressed

$$h = \frac{A_{pd}\eta(m+1)}{2\pi d^2} \cos^m(\phi_{tx}) T_s(\theta_{rx}) g(\theta_{rx}) \cos(\theta_{rx}), \quad (2)$$

where several key parameters and relationships are incorporated in the above equation: ϕ_{tx} represents the angle of irradiance, θ_{rx} signifies the angle of incidence surface,

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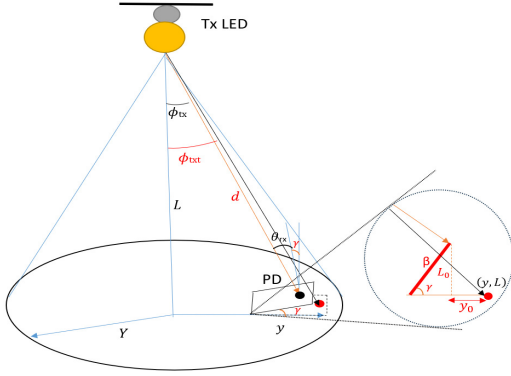


Fig. 1. Visible light positioning system model.

$T_s(\theta_{rx})$ stands for the gain associated with the receiver's optical filter, $g(\theta_{rx}) = \frac{n^2}{\sin^2(\phi_{FOV})}$ describes the gain of the optical concentrator, which relies on the refractive index n of the concentrator and the field-of-view (FOV) angle, i.e., ϕ_{FOV} . It's important to note that $g(\theta_{rx}) = 0$ when $\theta_{rx} > \phi_{FOV}$. Additionally, A_{pd} denotes the detection area of the photodetector and η represents the average responsivity of the receiver. Furthermore, $m = \frac{-1}{\log_2(\cos(\Phi_{1/2}))}$ is the Lambertian radiation order and is determined by the LED emission semi-angle $\Phi_{1/2}$ where the receiver and transmitter face each other $\cos(\phi_{tx}) = \cos(\theta_{rx})$.

Even more, we delve into scenarios where the receiver plane is not parallel to the transmitter plane, specifically addressing cases where the receiver is inclined at an angle γ [12], as illustrated in Fig. 1. In this general case where there is a tilting angle, the channel equation changed as $\cos(\theta_{rx}) = \cos(\phi_{tx} - \gamma) = \frac{L_t}{d} \cos \gamma + \frac{y_t}{d} \sin \gamma$ [13], where $\cos(\phi_{tx}) = \frac{L_t}{d}$, ϕ_{tx} is the angle of irradiation in case of having a tilting angle, d is the Euclidean distance from the transmitter to the receiver, which can be calculated as $d = \sqrt{L_t^2 + y_t^2}$, where L_t corresponds to the vertical distance from the LED to the PD surface and y_t is the horizontal distance from the center of the cell to the PD at any tilting angle value. Note that $L_t = L - L_0 = L - \beta \sin \gamma$, $y_t = y - y_0 = y - \beta(1 - \cos \gamma)$, where L and y are the vertical and horizontal distances at $\gamma = 0$ respectively, β is the PD length and L_0 and y_0 are the change in vertical and horizontal distances due to the presence of the tilting angle, as illustrated in Fig. 1, see [12] and the references therein. Hence, the received signal in (1) can be rewritten as

$$r = cxd^{-\alpha}x + \sqrt{cxd^{-\alpha}}n_{ds} + n, \quad (3)$$

where $\alpha = m + 3$ and

$$c = \frac{A_{pd}\eta(m+1)}{2\pi} T_s(\theta_{rx_t}) g(\theta_{rx_t}) L_t^m (L_t \cos \gamma + y_t \sin \gamma). \quad (4)$$

It's worth noting that the distortion variance is directly influenced by the distance between the transmitter and the receiver. This sets the additive distortion term apart from conventional receiver noise, which remains independent of distance.

III. RANGE ESTIMATION TECHNIQUES

In this section, we estimate the range d between the transmitter and the PD using two approaches. First, we use ML and

NLS estimators non-recursively. Then, we apply the EKF as a recursive method, allowing dynamic, time-adjusted estimates through iterative equations [14].

A. Maximum Likelihood Estimator

The ML estimator identifies unknown parameters by maximizing their likelihood function, as represented in their probability density function (PDF) [15]. We assume that the vertical distance (height) is a known quantity. Then, the vector of received signals, denoted by $\mathbf{r} = [r_1, r_2, \dots, r_N]^T$, where $(\cdot)^T$ represents the transpose operator, can be written as

$$\mathbf{r} = c\mathbf{x}d^{-\alpha} + \sqrt{cd^{-\alpha}}\mathbf{z} \odot \mathbf{n}_{ds} + \mathbf{n}. \quad (5)$$

Let N be the number of pilot symbols and $\mathbf{x} = [x_1, x_2, \dots, x_N]^T$ be the transmitted pilots' vector, and $\mathbf{z} = [\sqrt{x_1}, \sqrt{x_2}, \dots, \sqrt{x_N}]^T$ and $\odot$ is the Hadamard product operator and $\mathbf{n} = [n_1, n_2, \dots, n_N]^T$ is the thermal noise vector, and $\mathbf{n}_{ds} = [n_{ds1}, n_{ds2}, \dots, n_{dsN}]^T$ represents ISDSN vector. The elements of $\mathbf{n}$ are independent and identically distributed (i.i.d) random variables, i.e., $\mathbf{n} \sim \mathcal{N}(0, \sigma_n^2 \mathbf{I}_N)$ where $\mathbf{I}_N$ is the identity matrix of size N . The same for $\mathbf{n}_{ds}$, all the elements are i.i.d, so $\mathbf{n}_{ds} \sim \mathcal{N}(0, \zeta^2 \sigma_n^2 \mathbf{I}_N)$. It's important to note that all pilot symbols should hold positive values, i.e., $x_i > 0$, where $i \in \{1, 2, \dots, N\}$. The first step in deriving the ML estimator for our problem involves formulating the PDF for the initially received signal which can be expressed as

$$f(\mathbf{r}|d) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma_s^2}} \exp\left(-\frac{(r_i - cx_id^{-\alpha})^2}{2\sigma_s^2}\right). \quad (6)$$

The range estimation can be calculated by solving for $\frac{\partial}{\partial d} \ln f(\mathbf{r}|d) = 0$ as follows

$$\begin{aligned} \frac{\partial}{\partial d} \ln f(\mathbf{r}|d) &= \sum_{i=1}^N \frac{-1}{2\sigma_s^2} \frac{\partial \sigma_s^2}{\partial d} - \sum_{i=1}^N x_i c \alpha d^{-\alpha-1} \frac{(r_i - cx_id^{-\alpha})}{\sigma_s^2} \\ &+ \sum_{i=1}^N \frac{1}{2} (r_i - cx_id^{-\alpha})^2 (\sigma_s^2)^{-2} \frac{\partial \sigma_s^2}{\partial d} = 0, \end{aligned} \quad (7)$$

where $\sigma_s^2 = \sigma_n^2 + \sigma_{ds}^2$, recalling that $\sigma_{ds}^2 = x_i c d^{-\alpha} \zeta^2 \sigma_n^2$, therefore, $\frac{\partial \sigma_s^2}{\partial d} = -x_i c \alpha d^{-\alpha-1} \zeta^2 \sigma_n^2$.

It can be seen that finding a closed-form solution for the roots of the given equation $\hat{d}_{ML}$ manually is virtually impossible. However, numerical methods and software tools such as Matlab can easily compute the solution for $\hat{d}_{ML}$. Analytically studying the mean square error (MSE), specifically the distance error of the ML estimation, proves to be exceedingly challenging. Therefore, we rely on the simulation results presented later in the numerical section.

B. Nonlinear Least Square

Nonlinear least squares (NLS) problems arise when the received signal model cannot be expressed linearly in terms of the parameter we need to estimate like the relation between $\mathbf{r}$ and d as shown in (5). To simplify the NLS problems, we transform the parameters to linearize the problem [16]. By substituting $cd^{-\alpha}$ with g and solving it as a linear least square problem. The least-square estimator $\hat{g}_{LS}$ minimizes the sum of the squared differences between the observed and

predicted values [14]. Therefore, the least squares estimator can be calculated using (5) as

$$\hat{g}_{LS} = \arg \min_g \|\mathbf{r} - \mathbf{x}g\|^2 = \frac{\mathbf{x}^T \mathbf{r}}{\|\mathbf{x}\|^2}. \quad (8)$$

From (5), $\hat{g}_{LS}$ can be written as

$$\hat{g}_{LS} = \frac{\mathbf{x}^T \mathbf{x}}{\|\mathbf{x}\|^2} g + \frac{\mathbf{x}^T}{\|\mathbf{x}\|^2} \sqrt{g} \mathbf{z} \odot \mathbf{n}_{ds} + \frac{\mathbf{x}^T}{\|\mathbf{x}\|^2} \mathbf{n}. \quad (9)$$

The performance of the LS estimator can be evaluated by determining the MSE, which is equivalent to the estimation error variance. After some mathematical manipulations, the MSE of the estimator estimator can be given as

$$\sigma_{\hat{g}_{LS}}^2 = \frac{\sigma_n^2 \sum_{i=1}^N (1 + \sigma_{gds}^2)}{\|\mathbf{x}\|^2}, \quad (10)$$

where $\sigma_{gds}^2 = x_i g \zeta^2$. From (8), the estimated range $\hat{d}_{LS}$ can be written as

$$\hat{d}_{LS} = \left(\frac{\mathbf{x}^T \mathbf{r}}{c \|\mathbf{x}\|^2} \right)^{\frac{-1}{\alpha}}. \quad (11)$$

The MSE of the range estimator can be given as

$$\sigma_{\hat{d}_{LS}}^2 = \left(\frac{\partial \hat{d}}{\partial g} \right)^2 \sigma_{\hat{g}_{LS}}^2, \quad (12)$$

therefore If we assume the pilots are identical $x_i = p$, then the MSE can be simplified to

$$\sigma_{\hat{d}_{LS}}^2 = \frac{d^{2\alpha+2} \sigma_n^2 + p c d^{\alpha+2} \zeta^2 \sigma_n^2}{N p^2 c^2 \alpha^2}. \quad (13)$$

From (13), we observe that the MSE of the NLS estimator increases with rising ISDSN, a trend that will be explained in detail in the Simulation Results and Discussion section.

C. Extended Kalman Filter

To improve the precision of our range estimation over time, an initial estimation of d can be achieved using an ML estimator or assumed, followed by refinement through the application of EKF sequences. This process incorporates the assumption that the range follows a dynamic model influenced by time and is subject to uncertainty noise.

Due to its recursive nature, EKF's computational complexity is higher than that of ML and NLS, requiring multiple iterations and extensive calculations to update state estimates. The detailed computational complexity of EKF is discussed in [17], [18], [19].

The current range estimation $d[t]$ at time t can be determined by a linear transformation of its previous state $d[t-1]$ with added uncertainty noise $u[t]$ where $t > 0$ and is mathematically represented by

$$d[t] = d[t-1] + u[t]. \quad (14)$$

Furthermore, the dynamic behavior of the range is characterized by a Gaussian transition model, while the uncertainty noise is treated as a stochastic Gaussian process [15], [20]. Thus, $u(t)$ follows the Gaussian distribution with zero mean and variance σ_u^2 , i.e., $u[t] \sim N(0, \sigma_u^2)$. Moreover, the conditional probability of $f(d[t]|d[t-1]) \sim N(d[t-1], \sigma_u^2)$,

when we assume samples are independent. Additionally, the measurement or observation vector $\mathbf{r}[t]$ with dimension $N \times 1$ in (5), can be expressed as

$$\mathbf{r}[t] = \mathbf{e}(d[t]) + \mathbf{n}_s[t], \quad (15)$$

where $\mathbf{e}(d[t]) = c x d[t]^{-\alpha}$ and $\mathbf{n}_s[t] = \sqrt{c d[t]^{-\alpha}} \mathbf{z} \odot \mathbf{n}_{ds}[t] + \mathbf{n}[t]$. It is worth mentioning that $\mathbf{e}(d[t])$ represents the nonlinear equation modeling the relationship between the received vector and distance. Additionally, $\mathbf{n}_s[t]$ indicates the aggregate noise from thermal noise and shot noise, modeled as Gaussian with $\mathbf{n}_s[t] \sim \mathcal{N}(0, \mathbf{R})$, where $\mathbf{R} = \sigma_s^2 \mathbf{I}_N$.

When addressing the non-linear relationship between the received vector and the distance between the transmitter and receiver, we employ the EKF for its ability to manage non-linearity effectively [14]. The EKF method involves approximating the nonlinear $\mathbf{e}(d[t])$ function through a Taylor series expansion, specifically linearizing it at the point of the current estimate. This process allows the filter to update its state estimates by effectively handling the non-linearity of the system by linearizing $\mathbf{e}(d[t])$ around $\hat{d}[t-1]$ as follows.

$$\mathbf{e}(d[t]) \approx \mathbf{e}(\hat{d}[t-1]) + \left. \frac{\partial \mathbf{e}}{\partial d[t]} \right|_{d[t]=\hat{d}[t-1]} (d[t] - \hat{d}[t-1]). \quad (16)$$

We assume that the relationship between the received vector and range can be linearized, under the condition that noise is absent. This allows us to disregard the linearization effects of dependent noise. However, the influence of dependent noise is considered within the context of the covariance of measurement noise.

This approach enables a simplified linear analysis while ensuring that the complexities introduced by noise dependencies are accounted for in the uncertainty quantification of the measurements. Therefore, the measurement vector after linearization can be written as

$$\mathbf{r}[t] = \mathbf{s}[t]d[t] + \mathbf{n}_s[t] + \left(\mathbf{e}(\hat{d}[t-1]) - \mathbf{s}[t]\hat{d}[t-1] \right), \quad (17)$$

where

$$\mathbf{s}[t] = \left. \frac{\partial \mathbf{e}}{\partial d[t]} \right|_{d[t]=\hat{d}[t-1]}. \quad (18)$$

The following set of equations delineates the recursive sequence of steps necessary for the effective implementation of EKF, capturing the essence of the process required for its operation [15]:

$$\hat{d}[t-1] = \hat{d}[t-1|t-1], \quad (19a)$$

$$v[t-1] = v[t-1|t-1] + \sigma_u^2, \quad (19b)$$

$$\mathbf{k}[t] = v[t-1] \mathbf{s}^T[t] (\mathbf{R}[t] + \mathbf{s}[t]v[t-1] \mathbf{s}^T[t])^{-1}, \quad (19c)$$

$$\hat{d}[t] = \hat{d}[t-1] + \mathbf{k}[t](\mathbf{r}[t] - \mathbf{e}(\hat{d}[t-1])), \quad (19d)$$

$$v[t] = (1 - \mathbf{k}[t] \mathbf{s}[t])v[t-1], \quad (19e)$$

where the Kalman gain $\mathbf{k}[t]$ with dimension $1 \times N$, $v[t]$ is the minimum prediction MSE with dimension 1×1 . The initial value $\hat{d}[-1|-1]$ is set to d_0 , which can either be estimated from

the ML estimator or assumed. Through iterations, its effect diminishes. Moreover, the initial value of $v = [-1 | 1] = \sigma_u^2$.

IV. THEORITICAL BOUNDS: BAYESIAN CRAMÉR-RAO LOWER BOUND

In this section, we derive BCRB which represents the highest level of accuracy attainable in estimating the range $d[t]$ by any estimation techniques/algorithms. Initially, we derive the Fisher Information for the measurements $\mathbf{J}^M(d[t])$ as

$$\mathbf{J}^M(d[t]) = -\mathbb{E} \left\{ \frac{\partial^2}{\partial d^2[t]} \ln f(\mathbf{r}[t]|d[t]) \right\}, \quad (20)$$

where $\mathbb{E}$ is the expectation operator. Without loss of generality, and assuming that all the transmitted symbols are identical $x_i = p$ and after tedious mathematical manipulation the Fisher Information can be expressed as

$$\mathbf{J}^M(d[t]) = \frac{Np^2 c^2 \zeta^4 \alpha^2 (d[t])^{-2\alpha-2}}{2(1 + \zeta^2 c p (d[t])^{-\alpha})^2} + \frac{Np^2 c^2 \alpha^2 (d[t])^{-2\alpha-2}}{\sigma_n^2 + \sigma_n^2 \zeta^2 c (d[t])^{-\alpha} p}. \quad (21)$$

Subsequently, we obtain the Bayesian information $\mathbf{J}^B d[t]$ by amalgamating the insights derived from measurement data with prior knowledge as follows

$$\begin{aligned} \mathbf{J}^B d[t] &= \underbrace{\mathbf{J}^M(d[t])}_{\text{Measurements}} \\ &+ \underbrace{q_{22}[t] - q_{21}[t] [\mathbf{J}^B(d[t-1]) + q_{11}[t]]^{-1} q_{12}[t]}_{\text{Prior-Information}}. \end{aligned} \quad (22)$$

after detailed mathematical manipulations as discussed in [21], the quantities $q_{11}[t]$, $q_{12}[t]$, $q_{21}[t]$, and $q_{22}[t]$ can be derived and obtained as follows

$$q_{11}[t] = \mathbb{E} \left\{ -\frac{\partial^2 \ln f(d[t]|d[t-1])}{\partial d^2[t-1]} \right\} = \frac{1}{\sigma_u^2}, \quad (23a)$$

$$q_{21}[t] = \mathbb{E} \left\{ -\frac{\partial^2 \ln f(d[t]|d[t-1])}{\partial d[t-1] \partial d[t]} \right\} = \frac{-1}{\sigma_u^2}, \quad (23b)$$

$$q_{12}[t] = \mathbb{E} \left\{ -\frac{\partial^2 \ln f(d[t]|d[t-1])}{\partial d[t] \partial d[t-1]} \right\} = \frac{-1}{\sigma_u^2}, \quad (23c)$$

$$q_{22}[t] = \mathbb{E} \left\{ -\frac{\partial^2 \ln f(d[t]|d[t-1])}{\partial d^2[t]} \right\} = \frac{1}{\sigma_u^2}, \quad (23d)$$

where $\mathbf{J}^B(d[t-1])$ is the Bayesian Fisher information at the previous time. Furthermore, the BCRLB($d[t]$) of the estimation of $d[t]$ at the t^{th} sample is given by $\text{BCRLB}(d[t]) = 1/\mathbf{J}^B(d[t])$.

In considering the effect of ISDSN on the BCRLB, several key insights can be derived:

- *Remark 1:* The value of the measurement Fisher information $\mathbf{J}^M(d[t])$ increases as the number of pilots N increases, as shown in (21). This implies a reduction in BCRLB when N increases.
- *Remark 2:* As ζ tends towards zero, indicating no ISDSN, (21) can be expressed as

$$\lim_{\zeta \rightarrow 0} 1/\mathbf{J}^M(d[t]) = \frac{\sigma_n^2}{Np^2 c^2 \alpha^2 d[t]^{-2\alpha-2}}. \quad (24)$$

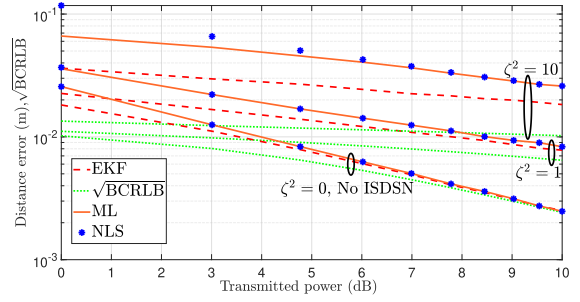


Fig. 2. Comparing the distance error of ML, NLS, and EKF estimators and $\sqrt{\text{BCRLB}}$ at different ζ^2 values.

This shows that when $\zeta = 0$, the BCRLB increases as the thermal noise σ_n^2 increases. However, increasing the number of pilots and transmitted power will decrease the BCRLB.

- *Remark 3:* From (21), it's evident that $\mathbf{J}^M(d[t])$ depends on the parameter we look to estimate, $d[t]$. Therefore, BCRLB depends on $d[t]$, with lower BCRLB when $d[t]$ is closer to the LED.
- *Remark 4:* At very high transmitted power, i.e., $p \rightarrow \infty$, $\mathbf{J}^M(d[t])$ is reduced to

$$\lim_{p \rightarrow \infty} 1/\mathbf{J}^M(d[t]) = \frac{\zeta^2 \sigma_n^2}{Np c \alpha^2 d[t]^{-\alpha-2}}. \quad (25)$$

It can be observed from (25) that BCRLB doesn't saturate at a high power level. Additionally, it is worth mentioning that $\text{BCRLB} \rightarrow 0$, when $p \rightarrow \infty$.

- *Remark 5:* It can be noticed from (24) that in the absence of ISDSN, BCRLB is proportional to $1/p^2$, while in the presence of ISDSN in (25), it is proportional to $1/p$. Therefore, ISDSN mitigates the improvement in BCRLB that can be received by increasing the transmitted power p .
- *Remark 6:* The estimated range $\hat{d}$ without ISDSN can be calculated using the ML estimator by substituting $\zeta = 0$ in (7), and thus it can be written as

$$\hat{d}^{-\alpha} = \frac{1}{c} \sum_{i=1}^N \frac{r_i}{x_i}. \quad (26)$$

This result is widely acknowledged in the literature, affirming the validity of our work.

V. SIMULATION RESULTS AND DISCUSSION

This section presents the numerical performance of SISO-VLP systems under ISDSN for range estimation between transmitter and PD using ML, NLS, and EKF techniques. Simulations are conducted in MATLAB, with results derived from approximately 10^6 Monte Carlo iterations per figure.

Furthermore, the simulation parameters are given as $\phi_{FOV} = 70^\circ$, $L = 3$ m, $A_{pd} = 0.1 \text{ m}^2$, $T_s(\theta_{rx}) = 1$, $g(\theta_{rx}) = 1$ and $\eta = 0.4$ A/Watt. Additionally, we assume $\sigma_n^2 = 10^{-2}$, $\sigma_u^2 = 10^{-3}$ and $\gamma = 0$ unless otherwise specified.

In Fig. 2, the root mean square error (RMSE) for the estimated distance $\hat{d}$ is plotted against the transmitted power from the source. This Figure illustrates the impact of ISDSN on ML, NLS, and EKF estimation methods, as well as the $(\sqrt{\text{BCRLB}})$.

When the ISDSN noise variance ζ^2 increases, the RMSE for all estimation methods rises accordingly. Notably, the

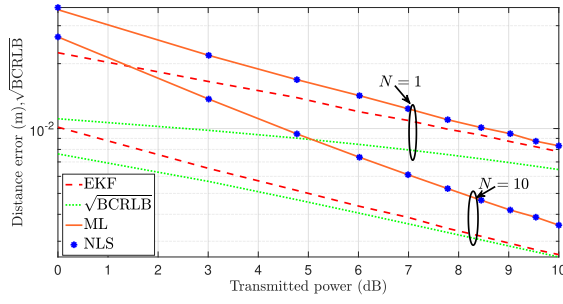


Fig. 3. $\sqrt{\text{BCRLB}}$ and distance error of the ML, NLS, and EKF estimators at different numbers of pilots for $\zeta^2 = 1$.

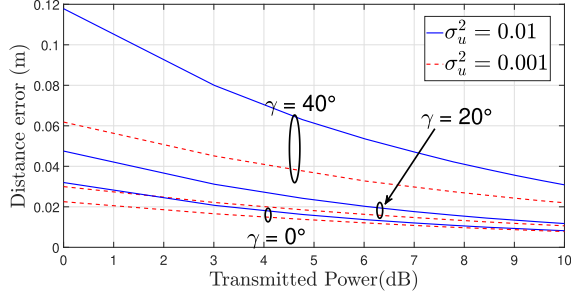


Fig. 4. The impact of tilting angle on the RMSE of the EKF estimator for $\zeta^2 = 1$.

$\sqrt{\text{BCRLB}}$ values also increase with increasing ζ^2 , consistent with the inverse of (21). Furthermore, the EKF demonstrates superior performance in terms of error reduction compared to the ML and NLS estimators, particularly at high ζ^2 values. Moreover, when $\zeta^2 = 0$, the RMSE of the estimator decreases with increasing power to align with $\sqrt{\text{BCRLB}}$. At high power levels, when $\zeta^2 = 0$ the variance of the NLS estimator becomes equal to the BCRLB this behavior is consistent with (13) and (24). It is important to note that from this figure the EKF has a higher time complexity compared to the other estimators because it is a recursive technique. However, it offers better performance at high ISDSN values, effectively mitigating the noise. For small signal-dependent noise values and to reduce complexity, the NLS or ML estimators can be used, as they provide RMSE values close to those of the EKF.

Fig. 3 discusses the effect of increasing the number of pilots N on the performance of ML, NLS, EKF estimators and $\sqrt{\text{BCRLB}}$, where the value of $\zeta^2 = 1$. The figure clearly shows that as the number of pilots grows, the error in distance estimation decreases, which is compatible with the mathematical results. From this figure, we can see that at high power and a large number of pilots EKF RMSE and $\sqrt{\text{BCRLB}}$ become the same, which is not the case in ML and NLS estimators. This finding can be explained by the fact that EKF tracks the range of a sequence of time iterations and these prior measurements enhance the performance of EKF.

Fig. 4 examines the impact of the receiver's tilting angle and uncertainty noise on the performance of the EKF estimator at $\zeta^2 = 1$, though this analysis is applicable for any value of ζ . The figure demonstrates that as the tilting angle of the receiver rises, the error of the estimated distance also rises. Similarly, when the uncertainty noise variance σ_u^2 grows, the distance error also grows. This rise in error due to the tilting angle occurs because the receiver plane does not face the transmitter,

reducing the performance of the estimation and decreasing the amount of received signal, consistently with (3) and (4).

VI. CONCLUSION

This study estimates user range using EKF, ML, and NLS, with EKF showing superior performance. As ISDSN noise decreases, the estimators converge and align with the BCRLB.

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